

$$\nu \frac{\partial T}{\partial y} = \frac{U_\infty}{2x} \sqrt{\frac{x\nu}{U_\infty}} (\eta f' - f) \cdot \frac{\partial T}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \sqrt{\frac{\nu}{U_\infty}}$$

$$\nu \frac{\partial T}{\partial y} = \frac{U_\infty}{2x} (\eta f' T' - f T')$$

Now for the right hand side:

$$\propto \frac{\partial^2 T}{\partial y^2} = \propto \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial y} \right) = \propto \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right) = \propto \frac{\partial \eta}{\partial y} \left(\frac{\partial}{\partial \eta} \left(\frac{\partial T}{\partial \eta} \right) \cdot \frac{\partial \eta}{\partial y} \right)$$

$$\propto \frac{\partial^2 T}{\partial y^2} = \propto \left(\frac{\partial \eta}{\partial y} \right)^2 \cdot T'' = \frac{U_\infty \alpha}{\nu x} T''$$

And the viscous term:

$$\frac{\nu}{c_p} \left(\frac{\partial u}{\partial y} \right)^2 = \frac{\nu}{c_p} \cdot \left(\frac{\partial u}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right)^2 = \frac{\nu}{c_p} \left(U_\infty f'' \cdot \sqrt{\frac{U_\infty}{\nu x}} \right)^2$$

$$\frac{\nu}{c_p} \left(\frac{\partial u}{\partial y} \right)^2 = \frac{\cancel{\nu} U_\infty^3}{\cancel{\nu} c_p x} (f'')^2$$

Putting it all together:

$$-\frac{\cancel{U_\infty \eta}}{2x} f' T' + \frac{\cancel{U_\infty \eta}}{2x} f' T' - \frac{\cancel{U_\infty}}{2x} f T' = \frac{\cancel{U_\infty} \alpha}{\cancel{\nu} x} T'' + \frac{U_\infty^3}{c_p x} (f'')^2$$

$\frac{\alpha}{\nu} = \frac{1}{Pr}$

$$\frac{1}{Pr} T'' + \frac{U_\infty^2}{c_p} (f'')^2 + \frac{1}{2} f T' = 0$$

$$\boxed{T'' + \frac{U_\infty^2 Pr}{c_p} (f'')^2 + \frac{1}{2} Pr f T' = 0} \quad (1)$$

So now we need to solve equation (1). Note, we can see that the equation is linear and that we have already solved for f before (Blasius).

$$\text{Let } \theta(\eta) = \frac{T - T_\infty}{U_\infty^2 / 2 c_p} \quad (2)$$

Substituting ② into ①

$$\boxed{\Theta'' + \frac{1}{2} Pr f \Theta' + 2 Pr (f'')^2 = 0} \Rightarrow \text{Second order, linear.} \quad \textcircled{3}$$

Using Duhamel's theorem, we can solve this by superposition of a particular and homogeneous solution.

$$\Theta(\eta) = \Theta_p(\eta) + \Theta_c(\eta)$$

For the particular solution; let's assume adiabatic wall conditions: $\Rightarrow$ We can do this due to Duhamel's theorem (Linear T)

$$\Theta_p(\eta) \Rightarrow \left. \frac{\partial \Theta_p}{\partial \eta} \right|_0 = 0, \quad \Theta_p(\eta \rightarrow \infty) = 0$$

To solve we can use the integrating factor method or solve numerically using a shooting scheme:

$$\Theta_p(\eta=0) = \Theta_{AW}(\eta=0) = \int_0^\infty \frac{\int_0^\eta \exp\left(\int_0^\eta \frac{1}{2} Pr f d\eta\right) 2 Pr (f'')^2 d\eta}{\exp\left(\int_0^\eta \frac{1}{2} Pr f d\eta\right)} d\eta$$

We can numerically integrate the above solution for various Pr and using $f(\eta)$ solution from Blasius (Tabulated result)

$$\boxed{\Theta_p(0) = \begin{cases} Pr^{1/2} & ; \quad 0.5 \leq Pr \leq 47 \\ 1.9 Pr^{1/3} & ; \quad Pr \geq 47 \end{cases}} \Rightarrow \text{Numerical solution} \\ \Rightarrow \text{Note, only solution at wall. Not full solution.}$$

So for the adiabatic wall:

$$\Theta_p(0) = \Theta_{AW}(0) = \frac{T_{AW} - T_\infty}{\frac{1}{2} U_\infty^2 / c_p} = r \equiv \text{recovery factor}$$

$$\boxed{T_{AW} = T_\infty + r \frac{U_\infty^2}{2c_p}}, \quad \boxed{r = \begin{cases} Pr^{1/2} & ; \quad 0.5 \leq Pr < 47 \\ 1.9 Pr^{1/3} & ; \quad Pr \geq 47 \end{cases}}$$

So the recovery factor represents the fraction of kinetic energy "recovered" by an adiabatic wall.

Note, let's check the limit of the solution: for $Pr = 1$

$$r = 1 \Rightarrow T_{AW} = T_{\infty} + \frac{U_{\infty}^2}{2C} \Rightarrow \text{Same as isentropic solution (stagnation point).}$$

Now we need the homogeneous solution

$$\Theta_{\#}(\eta) \Rightarrow \Theta_{\#}(\eta=0) = \text{constant (constant wall temp.)}$$

$$\Theta_{\#}(\eta \rightarrow \infty) = 0 ; \quad \boxed{\Theta_{\#}'' + \frac{1}{2} Pr f \Theta_{\#}' = 0} \quad (4)$$

$$\Theta_{\#} = \frac{T - T_0}{T_{\infty} - T_0} \Rightarrow \text{pg. (55)}$$

$\hookrightarrow$ Homogeneous Equation

So :

$$\Theta(\eta) = \Theta_p + C_1 \Theta_{\#} + C_2 \Rightarrow \text{Back substitute into (3) to check.}$$

Aside: Remember the following, for a nonhomogeneous linear ODE

$$y'' + p(t)y' + q(t)y = g(t)$$

$$y = y_c + y_p$$

Where y_p $\equiv$ any particular (specific) solution that satisfies the nonhomogeneous equation

$y_h \equiv C_1 y_1 + C_2 y_2$ is the general solution to the homogeneous equation: (complementary sol'n)

y_1 & y_2 are linearly independent solutions.

$$y'' + p(t)y' + q(t)y = 0 \Rightarrow \text{Homogeneous eqn.}$$

Back to our problem, we can now solve for C_1 & C_2 , using our overall b.c.'s

Just to be sure though, we can back substitute our assumed solution into (3) and see if it makes sense:

$$\Theta'' + \frac{1}{2} Pr f \Theta' + 2 Pr (f'')^2 = 0$$

$$\frac{\partial}{\partial \eta} \left(\frac{\partial}{\partial \eta} (\Theta_p + C_1 \Theta_H + C_2) \right) + \frac{1}{2} Pr f \frac{\partial}{\partial \eta} (\Theta_p + C_1 \Theta_H + C_2) + 2 Pr (f'')^2 = 0$$

$$\underbrace{\Theta_p'' + \frac{1}{2} Pr f \Theta_p' + 2 Pr (f'')^2}_{=0 \text{ (Complementary solution)}} + \underbrace{C_1 \Theta_H'' + \frac{1}{2} Pr f C_1 \Theta_H'}_{=0 \text{ (Homogeneous solution)}} = 0$$

Now back to our B.C.'s

$$\Theta(\eta \rightarrow \infty) = 0 \quad \text{or} \quad T(\eta \rightarrow \infty) = T_\infty$$

$$\Theta_H(\eta \rightarrow \infty) = \frac{T_\infty - T_0}{T_\infty - T_0} = 1$$

$$\Theta_p(\eta \rightarrow \infty) = 0 \quad (\text{From numerical solution})$$

$$0 = 0 + C_1(1) + C_2 \Rightarrow C_1 = -C_2$$

Our second b.c. is: $\Theta(\eta=0)$

$$\Theta_H(\eta=0) = \frac{T_0 - T_0}{T_\infty - T_0} = 0$$

$$\Theta_p(\eta=0) = \Theta_{Aw}(\eta=0) \quad (\text{Just a name change}) \quad \text{or } \Theta_p(0)$$

$$\frac{T_0 - T_\infty}{U_\infty^2 / 2 C_p} = \Theta_{Aw}(0) + 0 + C_2 \Rightarrow C_2 = \frac{T_0 - T_\infty}{U_\infty^2 / 2 C_p} - \Theta_{Aw}(0)$$

Back substituting everything into our complete solution

$$\Theta = \frac{T - T_\infty}{U_\infty^2 / 2 C_p} = \Theta_p + \left(\frac{T_0 - T_\infty}{U_\infty^2 / 2 C_p} - \frac{T_{0,Aw} - T_\infty}{U_\infty^2 / 2 C_p} \right) (1 - \Theta_H) \quad (110)$$

Rearranging, we obtain:

$$T - T_\infty = \Theta_p \frac{U_\infty^2}{2c_p} + (T_o - \overbrace{T_{o,AW}}^{T_{AW}})(1 - \Theta_H)$$

Now we can evaluate the heat flux (note, Θ_p still is solved numerically). Θ_H we solved before (pg. 55 of notes).

$$q''|_0 = -k \left. \frac{\partial T}{\partial y} \right|_0$$

$$\left. \frac{\partial T}{\partial y} \right|_0 = \left. \frac{\partial \Theta_p}{\partial y} \right|_0 \cdot \frac{U_\infty^2}{2c_p} - (T_s - T_{AW}) \left. \frac{\partial \Theta}{\partial y} \right|_0 ; \text{ note } \left. \frac{\partial \Theta_p}{\partial y} \right|_0 = 0 \text{ (Adiabatic)}$$

$$\left. \frac{\partial T}{\partial y} \right|_0 = 0 - (T_s - T_{AW}) \left. \frac{\partial \Theta}{\partial y} \right|_0$$

$$q''|_0 = -k \left. \frac{\partial T}{\partial y} \right|_s = k (T_s - T_{AW}) \left. \frac{\partial \Theta_H}{\partial \eta} \right|_0 \cdot \frac{\partial \eta}{\partial y}$$

$$q''|_0 = k (T_s - T_{AW}) \left. \frac{\partial \Theta_H}{\partial \eta} \right|_0 \cdot \sqrt{\frac{U_\infty}{\nu x}}$$

But we've already solved for $\Theta_H|_0$ on pg. 56

$$\Theta(\eta^*) = f'(\eta)$$

$$\frac{\partial \Theta_H}{\partial \eta} = \frac{\partial \Theta_H}{\partial \eta^*} \cdot \frac{\partial \eta^*}{\partial \eta} \Rightarrow \eta^* = \eta Pr^{1/3} \Rightarrow \frac{\partial \eta^*}{\partial \eta} = Pr^{1/3}$$

$$\left. \frac{\partial \Theta_H}{\partial \eta} \right|_0 = \left. \frac{\partial \Theta_H}{\partial \eta^*} \right|_0 \cdot Pr^{1/3} = f''(0) \cdot Pr^{1/3} = 0.332 Pr^{1/3}$$

$$\therefore \boxed{\left. \frac{\partial \Theta_H}{\partial \eta} \right|_0 = 0.332 Pr^{1/3}} \Rightarrow \text{Back substitute into } q''|_0$$

$$\boxed{q''|_0 = k (T_o - T_{AW}) \frac{0.332 Pr^{1/3}}{\sqrt{\nu x / U_\infty}}} \Rightarrow \text{Same result as before but } T_\infty = T_{AW}$$

So we can say for flows with viscous dissipation

So for viscous heating b.l.'s :

$$q''|_0 = h(T_0 - T_{AW})$$

$$h = k \frac{0.332 Pr^{1/3}}{\sqrt{Dx/U_\infty}} \Rightarrow \text{Same as before}$$

$$Nu_x = \frac{hx}{k} = 0.332 Pr^{1/3} Re_x^{1/2} \Rightarrow \text{Same as before}$$

The only change is instead of using T_∞ , we should use T_{AW} . Very cool!

Let's test the limits of this :

$$T_{AW} = T_\infty + r \frac{U_\infty^2}{2C_p}, \quad r = \begin{cases} Pr^{1/2} & ; 0.5 \leq Pr < 47 \\ 1.9 Pr^{1/3} & ; Pr \geq 47 \end{cases}$$

For $Pr=1$, $r=1$ so

$$T_{AW} = T_\infty + \frac{U_\infty^2}{2C_p} \Rightarrow \text{Same as previous solution (stagnation pt.)}$$

$$\text{For: } q''|_0 = h(T_0 - T_\infty - \frac{U_\infty^2}{2C_p}) \Rightarrow \frac{U_\infty^2}{2C_p(T_0 - T_\infty)} \ll 1$$

Then $T_{AW} = T_\infty$ and our old results are valid.

Here we see why we defined the Eckert number like we did:

$$Ec = \frac{U_\infty^2}{C_p(T_0 - T_\infty)} \Rightarrow \text{It's nothing but a measure of the negligibility of the recovery factor term.}$$

Note to be more rigorous, we should say:

$$Ec = \frac{r U_\infty^2}{2C_p(T_0 - T_\infty)} \ll 1 \text{ for } T_{AW} = T_\infty \text{ since } r = f(Pr)$$

Sometimes our condition is written as $Pr \cdot Ec \ll 1$

So how about our second condition (temperature dependent properties). Well, Eckert found a miraculous solution. As long as we take our properties at: (as long as $C_p \approx \text{const}$)

$$\boxed{T_R = T_\infty + 0.5(T_o - T_\infty) + 0.22(T_{Aw} - T_\infty)} \quad \text{for } Ma < 20, \quad Pr < 15.$$

All properties are evaluated at T_R , including Pr, ρ, μ, k . Note, this also takes into account compressibility effects. This also works for calculating shear stress & local skin friction coefficient. Use T_R as the reference temperature.

Example #1 / For turbulent flows, it's been shown that:

$$r \approx Pr^{1/3} \quad \text{for all } Pr, \text{ and } 0 < Ma < 8, \text{ and gas flow}$$

Find T_{Aw} for the space shuttle re-entry. ($Ma = 5$)

$$V = 5(343 \text{ m/s}) = 1715 \text{ m/s}$$

$$C_{p, \text{air}} \approx 1000 \text{ J/kg} \cdot \text{K}$$

$$Pr_{, \text{air}} \approx 0.68 \quad (\text{at } T = 400^\circ\text{C})$$

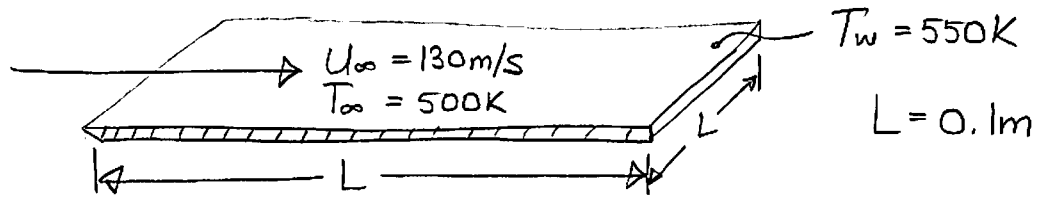
$$r = Pr^{1/3} = 0.88 \Rightarrow T_{Aw} = -40^\circ\text{C} + (0.88) \frac{(1715 \text{ m/s})^2}{2(1000 \text{ J/kg} \cdot \text{K})}$$

$$\boxed{T_{Aw} \approx 1254^\circ\text{C}}$$

Note, now we should iterate to get properties at T_R . C_p won't change that much, but Pr will.

Also, at very high gas velocities ($Ma > 20$), the wall gets so hot that ionization of the gas occurs, which needs to be taken into account. This ionization is why we lose radio communication with re-entry vehicles. The charged ions act as a blocking mechanism for EM waves. (113)

Example #2 | Flow past a flat plate (air)



First let's check to see if viscous heating is important:

$$Pr \cdot Ec = Pr \frac{U_\infty^2}{c_p \Delta T} = 0.23 \Rightarrow \text{Need to consider}$$

$$Re_L = 310\,000 < 5.0 \times 10^5 \text{ (Laminar)}$$

$$Ma = 0.28 \approx \text{incompressible}$$

Now we can solve since all the conditions fit our developed correlations.

$$T_{Aw} = T_\infty + \frac{r U_\infty^2}{2 c_p}, \quad r = Pr^{1/2} = (0.68)^{1/2} = 0.82$$

$$T_{Aw} = 500 \text{ K} + \frac{0.82 (130 \text{ m/s})^2}{2 (1000 \text{ J/kg} \cdot \text{K})} = 506.9 \text{ K}$$

$$T_R = T_\infty + 0.5 (T_o - T_\infty) + 0.22 (T_{Aw} - T_\infty) \\ = 500 \text{ K} + 25 \text{ K} + 0.22 (6.9 \text{ K})$$

$$T_R = 526.5 \text{ K} \Rightarrow \rho_{\text{Air}} = 0.65 \text{ kg/m}^3, \quad c_{p,\text{Air}} \approx 1050 \text{ J/kg} \cdot \text{K} \\ \mu_{\text{Air}} = 2.67 \times 10^{-5} \text{ kg/ms}, \quad Pr \approx 0.68, \quad k_{\text{Air}} = 0.04 \text{ W/mK}$$

$$q''|_0 = h L^2 (T_o - T_{Aw})$$

$$h = \overline{Nu}_L \cdot \frac{k_{\text{Air}}}{L} = \frac{k_{\text{Air}}}{L} \cdot (0.664 Re_L^{1/2} Pr^{1/3}) = 136.3 \text{ W/m}^2 \cdot \text{K}$$

$$\boxed{q''|_0 = 58.84 \text{ W}} \\ \text{heat}$$

$$\boxed{q''|_{0,\text{false}} = h A (T_o - T_\infty) = 68 \text{ W}}$$

$q''_{\text{real}} < q''_{\text{false}}$ due